

Product Requirements Document (PRD)

Project: AI-Assisted Estimation & Pricing Engine

Company: OneSyntax **Version:** 1.4 (Updated with Metrics Translation Engine)

1. Project Overview

The primary objective of this system is to generate highly accurate, risk-managed pricing quotes for software projects. This is **not** a fully autonomous AI system. AI is exclusively used to generate the Work Breakdown Structure (WBS) and flag hidden technical risks. All mathematical calculations, data conversions, and final decisions are strictly handled by deterministic code and Human-in-the-loop (Tech Lead) validation.

2. Core Principles

- **No AI Math:** The LLM will perform zero calculations (E, SD, RSD, Conversions). All math is handled by backend hard-coded formulas.
- **Human-in-the-loop (Crucial):** AI-generated WBS tasks and baseline estimates must be reviewed, modified (if necessary), and explicitly approved by a Tech Lead before proceeding.
- **Controlled Transparency:** Raw internal data (O, M, P values, Story Points, Standard Deviations) are completely hidden from the client. The system maintains transparency by sharing the detailed scope, business assumptions, and a calculated time/cost range.

3. System Architecture & Modules

Module 1: AI Work Breakdown Structure (WBS) Generator

- **Input:** Client Requirement Document or Project Brief.
- **Process (AI LLM):** Breaks down the project into granular technical tasks using the **INVEST framework** (Independent, Negotiable, Valuable, Estimable, Small, Testable) to ensure tasks are properly sized for estimation.
- **Validation:** Tech Lead reviews, edits, and approves the final task list.

Module 2: AI Risk Flagger

- **Process (AI LLM):** Analyzes each WBS task to identify edge cases, security vulnerabilities, API rate limits, or legacy code constraints.
- **Output:** "Risk Notes" attached to each task to help the Tech Lead accurately determine the Pessimistic (P) scenario.

Module 3: 3-Point Estimation Input (UI/UX)

- **Process:** Tech Lead inputs the following three values (in Story Points or Hours) for each task, guided by AI baseline suggestions:
 - Optimistic (O) - Best case
 - Most Likely (M) - Normal case
 - Pessimistic (P) - Worst case (incorporating Risk Notes)

Module 4: Deterministic Math Engine (Calculation)

- **Process:** Calculates PERT values using backend logic.
- **Formulas:**
 - Expected Value (E) = $(O + 4M + P) / 6$
 - Standard Deviation (SD) = $(P - O) / 6$
 - Relative Standard Deviation (RSD) = $(SD / E) * 100$

Module 5: Pricing Strategy Decision Engine

- **Logic based on RSD:**
 - **RSD < 10%:** [GREEN FLAG] Low Risk. Safe to offer a Fixed Price based on the Expected Value.
 - **RSD 10% - 20%:** [YELLOW FLAG] Medium Risk. Offer a Fixed Price but securely inject a Risk Buffer (e.g., Expected Value + 1 SD).
 - **RSD > 20%:** [RED FLAG] High Risk. Deny Fixed Price. System automatically recommends a Time & Materials (T&M) / Hourly contract.

Module 6: Client-Facing Proposal Generator (Safe Transparency)

- **Process:** Translates internal agile metrics into a professional, risk-protected client proposal.
- **Features & Outputs (What the client sees):**
 1. **Detailed Scope (WBS):** Displays the complete, approved task list to show the depth of work.
 2. **Assumptions & Exclusions:** Translates Module 2 Technical Risks into business conditions. (e.g., "Assumes full access to 3rd-party API documentation. Unforeseen API deprecations are excluded and subject to change.")
 3. **Metrics Translation Engine (Points to Time & Money):**
 - *Internal Inputs:* Team Velocity (Points/Week), Team Capacity (Hours/Week), Hourly Rate (\$).
 - *Calculation:* Dynamically converts total project points/SD into real-world metrics.
 - $\text{Hours per Point} = \text{Capacity} / \text{Velocity}$
 - $\text{Total Weeks} = \text{Total Points} / \text{Velocity}$
 - $\text{Total Cost} = \text{Total Hours} \times \text{Hourly Rate}$
 4. **Transparent Pricing Model:**
 - Displays a **Calculated Safe Range** (e.g., Cost: \$5,000 - \$6,500 | Timeline: 4 - 5 Weeks) derived from the +1 SD or +2 SD conversions.
 - *Disclaimer clause:* "These estimates are strictly based on the agreed Assumptions & Exclusions. If project scope or assumptions change, this range is subject to modification."

4. Technical Stack Considerations

- **Frontend:** React, Vue, or Svelte (for real-time calculation updates in the UI).
- **Backend:** Node.js / Python.
- **AI Integration:** OpenAI API (GPT-4o) or Anthropic Claude using strictly "Structured Outputs (JSON)" for WBS and Risk Flagging.

